

BHV_ConstantDepth, BHV_ConstHeading, and BHV_ConstantSpeed

Constant setpoint behaviors.

Included MIT MOOS-IvP PDF sources:

- bhv_const_depth.pdf
- bhv_const_hdg.pdf
- bhv_const_speed.pdf

The ConstantDepth Behavior

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1 The ConstantDepth Behavior

This behavior will drive the vehicle at a specified depth. This behavior merely expresses a preference for a particular depth. If other behaviors also have a depth preference, coordination/compromise will take place through the multi-objective optimization process. The following parameters are defined for this behavior:

1.1 Configuration Parameters

Listing 1.1: Configuration Parameters Common to All Behaviors.

<code>activeflag</code> :	A MOOS variable-value pair posted when the behavior is in the <i>active</i> state. [more] .
<code>condition</code> :	Specifies a condition that must be met for the behavior to be running. [more] .
<code>duration</code> :	Time in behavior will remain running before declaring completion. [more] .
<code>duration_idle_decay</code> :	When true, duration clock is running even when in the <i>idle</i> state. [more] .
<code>duration_reset</code> :	A variable-pair such as <code>MY_RESET=true</code> , that will trigger a duration reset. [more] .
<code>duration_status</code> :	The name of a MOOS variable to which the vehicle duration status is published. [more] .
<code>endflag</code> :	A MOOS variable-value pair posted when the behavior has completed. [more] .
<code>idleflag</code> :	A MOOS variable-value pair posted when the behavior is in the <i>idle</i> state. [more] .

<code>inactiveflag</code> :	A MOOS variable-value posted when the behavior is <i>not</i> in the <i>active</i> state. [more] .
<code>name</code> :	The (unique) name of the behavior. [more] .
<code>nostarve</code> :	Allows a behavior to assert a maximum staleness for a MOOS variable. [more] .
<code>perpetual</code> :	If true allows the behavior to run even after it has completed. [more] .
<code>post_mapping</code> :	Re-direct behavior output normally to one MOOS variable to another instead. [more] .
<code>priority</code> :	The priority weight of the behavior. [more] .
<code>pwt</code> :	Same as <code>priority</code> .
<code>runflag</code> :	A MOOS variable and a value posted when a behavior has met its conditions. [more] .
<code>spawnflag</code> :	A MOOS variable and a value posted when a behavior is spawned. [more] .
<code>spawnxflag</code> :	A MOOS variable and a value posted when a behavior is spawned. [more] .
<code>templating</code> :	Turns a behavior into a template for spawning behaviors dynamically. [more] .
<code>updates</code> :	A MOOS variable from which behavior parameter updates are read dynamically. [more] .

Listing 1.2: Configuration Parameters for the ConstantDepth Behavior.

Parameter	Description
<code>basewidth</code> :	The width of the base, in meters, in the produced ZAIC-style IvP function. Section 1.4.
<code>depth</code> :	The desired depth of the vehicle, in meters.
<code>duration</code> :	Behavior duration, in seconds. Mandatory configuration for this behavior. Section 1.3.
<code>peakwidth</code> :	The width of the peak, in meters, in the produced ZAIC-style IvP function. Section 1.4.
<code>summitdelta</code> :	The height of the summit delta parameter in the produced ZAIC-style IvP function. See Figure ?? . Section 1.4.
<code>depth_mismatch_var</code> :	Name of the MOOS variable indicating the present delta between the desired depth and the current depth. Section 1.2.

Listing 1.3: Example Configuration Block.

```

Behavior = BHV_ConstantDepth
{
  // General Behavior Parameters
  // -----
  name          = const_dep_survey  // example
  pwt           = 100                // default
  condition     = MODE==SURVEYING   // example
  updates       = CONST_DEP_UPDATES // example

  // Parameters specific to this behavior
  // -----
          basewidth = 100            // default
          depth     = 0              // default
  depth_mismatch_var = DEPTH_DIFF    // example
          duration  = 0              // default (Choose something higher!)
          peakwidth  = 3              // default
          summitdelta = 50           // default
}

```

1.2 Variables Published

The behavior may be optionally configured to publish a variable indicating the discrepancy between the requested depth and the actual observed depth. The MOOS variable is named using the `depth_mismatch_var` parameter. This variable will only be published when the behavior is in the running state.

1.3 The `duration` Parameter

The `duration` parameter defined for all general behaviors, but for this behavior, specification is mandatory for safety reasons. The default if not specified is 0 seconds which will result in the behavior completing immediately. If no duration limit is desired, e.g., if the behavior is tied to another behavior or event via condition variables, then setting `duration=no-time-limit` will result in no time duration checks for this behavior.

1.4 The ConstantDepth Objective Function

See Figure ??.

- `basewidth`: The width of the base, in meters in the produced objective function. The default is 100. See Figure ?? for more on the basewidth parameter used in the ZAIC tool for building IvP functions.
- `peakwidth`: The width of the peak in meters in the produced objective function. The default is 3. See Figure ?? for more on the peak parameter used in the ZAIC tool for building IvP functions.
- `summitdelta`: The width of the base, in meters in the produced objective function. The default is 50. See Figure ?? for more on the summitdelta parameter used in the ZAIC tool for building IvP functions.

The ConstantHeading Behavior

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1 The ConstantHeading Behavior

This behavior will drive the vehicle at a specified heading. This behavior merely expresses a preference for a particular heading. If other behaviors also have a heading preference, coordination/compromise will take place through the multi-objective optimization process.

1.1 Configuration Parameters

Listing 1.1: Configuration Parameters Common to All Behaviors.

<code>activeflag</code> :	A MOOS variable-value pair posted when the behavior is in the <i>active</i> state. [more] .
<code>condition</code> :	Specifies a condition that must be met for the behavior to be running. [more] .
<code>duration</code> :	Time in behavior will remain running before declaring completion. [more] .
<code>duration_idle_decay</code> :	When true, duration clock is running even when in the <i>idle</i> state. [more] .
<code>duration_reset</code> :	A variable-pair such as <code>MY_RESET=true</code> , that will trigger a duration reset. [more] .
<code>duration_status</code> :	The name of a MOOS variable to which the vehicle duration status is published. [more] .
<code>endflag</code> :	A MOOS variable-value pair posted when the behavior has completed. [more] .
<code>idleflag</code> :	A MOOS variable-value pair posted when the behavior is in the <i>idle</i> state. [more] .

<code>inactiveflag</code> :	A MOOS variable-value posted when the behavior is <i>not</i> in the <i>active</i> state. [more] .
<code>name</code> :	The (unique) name of the behavior. [more] .
<code>nostarve</code> :	Allows a behavior to assert a maximum staleness for a MOOS variable. [more] .
<code>perpetual</code> :	If true allows the behavior to run even after it has completed. [more] .
<code>post_mapping</code> :	Re-direct behavior output normally to one MOOS variable to another instead. [more] .
<code>priority</code> :	The priority weight of the behavior. [more] .
<code>pwt</code> :	Same as <code>priority</code> .
<code>runflag</code> :	A MOOS variable and a value posted when a behavior has met its conditions. [more] .
<code>spawnflag</code> :	A MOOS variable and a value posted when a behavior is spawned. [more] .
<code>spawnxflag</code> :	A MOOS variable and a value posted when a behavior is spawned. [more] .
<code>templating</code> :	Turns a behavior into a template for spawning behaviors dynamically. [more] .
<code>updates</code> :	A MOOS variable from which behavior parameter updates are read dynamically. [more] .

Listing 1.2: Configuration Parameters for the ConstantHeading Behavior.

Parameter	Description
<code>basewidth</code>	The width of the base, in degrees, in the produced ZAIC-style IvP function. Section 1.5.
<code>complete_thresh</code>	The threshold in discrepancy between the requested heading and present heading before the behavior completes. The default is -1 . Section 1.4.
<code>duration</code>	Behavior duration, in seconds. Mandatory configuration for this behavior. Section 1.3.
<code>heading</code>	The desired heading of the vehicle, in degrees (North=0).
<code>peakwidth</code>	The width of the peak, in degrees, in the produced ZAIC-style IvP function. Section 1.5.
<code>summitdelta</code>	The height of the summit delta parameter in the produced ZAIC-style IvP function. Section 1.5.
<code>heading_mismatch_var</code>	Name of the MOOS variable indicating the present delta between the desired heading and the current heading. Section 1.2.

Listing 1.3: Example Configuration Block.

```

Behavior = BHV_ConstantHeading
{
  // General Behavior Parameters
  // -----
  name          = const_hdg          // example
  pwt           = 100                // default
  condition     = MODE==GO_STRAIGHT // example
  updates       = CONST_HDG_UPDATES  // example

  // Parameters specific to this behavior
  // -----
          basewidth = 10              // default
          duration  = 0                // default
          heading   = 0                // default
  heading_mismatch_var = HDG_DIFF      // example
          peakwidth = 10              // default
          summitdelta = 25             // default
}

```

1.2 Variables Published

The behavior may be optionally configured to publish a variable indicating the discrepancy between the requested heading and the actual observed heading. The MOOS variable is named using the `heading_mismatch_var` parameter. This variable will only be published when the behavior is in the running state.

1.3 The `duration` Parameter

The `duration` parameter defined for all general behaviors, but for this behavior, specification is mandatory for safety reasons. The default if not specified is 0 seconds which will result in the behavior completing immediately. If no duration limit is desired, e.g., if the behavior is tied to another behavior or event via condition variables, then setting `duration = no-time-limit` will result in no time duration checks for this behavior.

1.4 Behavior Completion

The behavior, by default, remains active so long as its conditions are met, just like any other behavior. If the user optionally declares a threshold via the `complete_thresh` parameter, the behavior will complete once the discrepancy between the observed heading and the goal heading falls below that threshold. In this case the behavior will complete, as any other helm behavior, by posting its endflags.

1.5 The ConstantHeading Objective Function

See Figure ??.

- **basewidth**: The width of the base, in degrees in the produced objective function. The default is 170. See Figure ?? for more on the basewidth parameter used in the ZAIC tool for building IvP functions.

- **peakwidth**: The width of the peak in degrees in the produced objective function. The default is 10. See Figure ?? for more on the peak parameter used in the ZAIC tool for building IvP functions.
- **summitdelta**: The width of the base, in meters in the produced objective function. The default is 25. See Figure ?? for more on the summitdelta parameter used in the ZAIC tool for building IvP functions.

The ConstantSpeed Behavior

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Contents

1 The ConstantSpeed Behavior

This behavior will drive the vehicle at a specified speed. This behavior merely expresses a preference for a particular speed. If other behaviors also have a speed preference, coordination/compromise will take place through the multi-objective optimization process. The following parameters are defined for this behavior:

1.1 Configuration Parameters

Listing 1.1: Configuration Parameters Common to All Behaviors.

<code>activeflag</code> :	A MOOS variable-value pair posted when the behavior is in the <i>active</i> state. [more] .
<code>condition</code> :	Specifies a condition that must be met for the behavior to be running. [more] .
<code>duration</code> :	Time in behavior will remain running before declaring completion. [more] .
<code>duration_idle_decay</code> :	When true, duration clock is running even when in the <i>idle</i> state. [more] .
<code>duration_reset</code> :	A variable-pair such as <code>MY_RESET=true</code> , that will trigger a duration reset. [more] .
<code>duration_status</code> :	The name of a MOOS variable to which the vehicle duration status is published. [more] .
<code>endflag</code> :	A MOOS variable-value pair posted when the behavior has completed. [more] .
<code>idleflag</code> :	A MOOS variable-value pair posted when the behavior is in the <i>idle</i> state. [more] .

<code>inactiveflag</code> :	A MOOS variable-value posted when the behavior is <i>not</i> in the <i>active</i> state. [more] .
<code>name</code> :	The (unique) name of the behavior. [more] .
<code>nostrarve</code> :	Allows a behavior to assert a maximum staleness for a MOOS variable. [more] .
<code>perpetual</code> :	If true allows the behavior to run even after it has completed. [more] .
<code>post_mapping</code> :	Re-direct behavior output normally to one MOOS variable to another instead. [more] .
<code>priority</code> :	The priority weight of the behavior. [more] .
<code>pwt</code> :	Same as <code>priority</code> .
<code>runflag</code> :	A MOOS variable and a value posted when a behavior has met its conditions. [more] .
<code>spawnflag</code> :	A MOOS variable and a value posted when a behavior is spawned. [more] .
<code>spawnxflag</code> :	A MOOS variable and a value posted when a behavior is spawned. [more] .
<code>templating</code> :	Turns a behavior into a template for spawning behaviors dynamically. [more] .
<code>updates</code> :	A MOOS variable from which behavior parameter updates are read dynamically. [more] .

Listing 1.2: Configuration Parameters for the ConstantSpeed Behavior.

Parameter	Description
<code>basewidth</code>	The width of the base, in meters per second, in the produced ZAIC-style IvP function. Section 1.4.
<code>duration</code>	Behavior duration, in seconds. Mandatory configuration for this behavior. Section 1.3.
<code>peakwidth</code>	The width of the peak, in meters per second, in the produced ZAIC-style IvP function. Section 1.4.
<code>speed</code>	The desired speed of the vehicle, in meters per second.
<code>summitdelta</code>	The height of the summit delta parameter in the produced ZAIC-style IvP function. Section 1.4.
<code>speed_mismatch_var</code>	Name of the MOOS variable indicating the present delta between the desired speed and the current speed. If left unspecified, no posting is made. Section 1.2.

Listing 1.3: Example Configuration Block.

```

Behavior = BHV_ConstantSpeed
{
  // General Behavior Parameters
  // -----
  name          = const_spd_transit // example
  pwt           = 100               // default
  condition     = MODE==TRANSITING // example
  updates       = CONST_SPD_UPDATES // example

  // Parameters specific to this behavior
  // -----
          basewidth = 0.2           // default
          duration  = 0             // default
          speed     = 0             // default
  speed_mismatch_var = SPEED_DIFF // example
          peakwidth = 0             // default
          summitdelta = 0           // default
}

```

1.2 Variables Published

The behavior may be optionally configured to publish a variable indicating the discrepancy between the requested speed and the actual observed speed. The MOOS variable is named using the `speed_mismatch_var` parameter. This variable will only be published when the behavior is in the running state.

1.3 The `duration` Parameter

The `duration` parameter defined for all general behaviors, but for this behavior, specification is mandatory for safety reasons. The default if not specified is 0 seconds which will result in the behavior completing immediately. If no duration limit is desired, e.g., if the behavior is tied to another behavior or event via condition variables, then setting `duration = no-time-limit` will result in no time duration checks for this behavior.

1.4 The ConstantSpeed Objective Function

See Figure ??.

- `peakwidth`: The width of the peak in meters/second in the produced objective function. The default is 0. See Figure ?? for more on the peak parameter used in the ZAIC tool for building IvP functions.
- `basewidth`: The width of the base, in meters/second in the produced objective function. The default is 0.2. See Figure ?? for more on the basewidth parameter used in the ZAIC tool for building IvP functions.
- `summitdelta`: The width of the base, in meters/second in the produced objective function. The default is 0. See Figure ?? for more on the summitdelta parameter used in the ZAIC tool for building IvP functions.